

Proposition B.8. Suppose that each $p^0|_{C_k}$ is reversible with spectral gap $\gamma(\mathbf{P}_{kk}^0) \geq \gamma$ and the stationary measure $\mu_k := \pi_k^0$ satisfies $\rho/M \leq \mu_k(x) \leq \rho'/M$. Moreover suppose that the eigenvalue matrix $\mathbf{V}_k$ of $\mathbf{P}_{kk}^0$ has condition number bounded as $\kappa(\mathbf{V}_k) = \|\mathbf{V}_k\|_2 \|\mathbf{V}_k^{-1}\|_2 \leq \kappa_0 \sqrt{M}$, and the group inverse $\mathbf{A}_k^\#$ of $\mathbf{I} - \mathbf{P}_{kk}^0$ satisfies $\|\mathbf{A}_k^\#\|_\infty \leq g_0$ for constants κ_0, g_0 . Then for all $\varepsilon = o(M^{-1})$,

$$\gamma^\dagger(\mathbf{S}_{kk}^\varepsilon) \geq \frac{\gamma}{2} \quad \text{and} \quad \frac{\rho}{2M} \leq \pi_k^\varepsilon(x) \leq \frac{2\rho'}{M} \quad \forall x \in C_k.$$

Proof. By the proportionality of p^ε in Assumption 2 and (4) we have

$$\begin{aligned} \|\mathbf{S}_{kk}^\varepsilon - \mathbf{P}_{kk}^0\|_{1,\infty} &\leq \|\mathbf{S}_{kk}^\varepsilon - \mathbf{P}_{kk}^\varepsilon\|_{1,\infty} + \|\mathbf{P}_{kk}^\varepsilon - \mathbf{P}_{kk}^0\|_{1,\infty} \\ &\leq d_{\text{out}}\varepsilon + \max_{x \in C_k} \sum_{y \in C_k} |p^0(y|x) - p^\varepsilon(y|x)| \\ &\leq 2d_{\text{out}}\varepsilon, \end{aligned}$$

so that $\|\mathbf{S}_{kk}^\varepsilon - \mathbf{P}_{kk}^0\|_2 \leq 2\sqrt{M}d_{\text{out}}\varepsilon$ by Lemma B.5. Then by the Bauer-Fike theorem it holds that

$$|\lambda_2(\mathbf{S}_{kk}^\varepsilon) - \lambda_2(\mathbf{P}_{kk}^0)| \leq \kappa(\mathbf{V}_k) \|\mathbf{S}_{kk}^\varepsilon - \mathbf{P}_{kk}^0\|_2 \leq 2\kappa_0 M d_{\text{out}}\varepsilon = o(1),$$

therefore $\gamma^\dagger(\mathbf{S}_{kk}^\varepsilon) \geq \gamma(\mathbf{S}_{kk}^\varepsilon) \geq \frac{\gamma}{2}$ for sufficiently large M . Furthermore by the condition number bound in Meyer (1980) the perturbed stationary distribution satisfies

$$\|\pi_k^\varepsilon - \mu_k\|_\infty \leq \|\mathbf{A}_k^\#\|_\infty \|\mathbf{S}_{kk}^\varepsilon - \mathbf{P}_{kk}^0\|_\infty \leq 2g_0 d_{\text{out}}\varepsilon = o(M^{-1}),$$

proving the second assertion. $\square$

With these results in mind, we can prove the following concentration bound for the reduced chain $\tilde{X}^{k,\varepsilon}$.

Lemma B.9. For all $x, y \in C_k$ and $\delta > 0$ it holds that $\mathbb{P}_x(\tilde{\tau}_y^{k,\varepsilon} \geq m) \leq \delta$ as long as

$$m \geq \frac{8M}{\rho\gamma} \log \frac{1}{\delta} \cdot \log \frac{M}{\rho}. \quad (5)$$

Proof. By Proposition B.4, the mixing time of $\tilde{X}_t^{k,\varepsilon}$ is bounded above as

$$t_{\text{mix}} := t_{\text{mix}}\left(\frac{\rho}{2M}\right) \leq \frac{1}{\gamma^\dagger(\mathbf{S}_{kk}^\varepsilon)} \left(1 + 2 \log \frac{M}{\rho} + \log \frac{1}{\min \pi_k^\varepsilon}\right) \leq \frac{4}{\gamma} \log \frac{M}{\rho}$$

so that for any $x, y \in C_k$,

$$(s_{kk}^\varepsilon)^{t_{\text{mix}}}(y|x) \geq \pi_k^\varepsilon(y) - (\pi_k^\varepsilon(y) - (s_{kk}^\varepsilon)^{t_{\text{mix}}}(y|x)) \geq \frac{\rho}{M} - \|(s_{kk}^\varepsilon)^{t_{\text{mix}}} - \pi_k^\varepsilon\|_{\text{TV}} \geq \frac{\rho}{2M}.$$

This implies each step of the t_{mix} -skipped chain $(\tilde{X}_{t_{\text{mix}}t}^{k,\varepsilon})_{t \geq 0}$ is well-mixed, and hence

$$\begin{aligned} \mathbb{P}_x(\tilde{\tau}_y^{k,\varepsilon} \geq m) &\leq \sup \prod_{t=1}^{\lfloor m/t_{\text{mix}} \rfloor} \left(1 - \mathbb{P}_{\tilde{X}_{t_{\text{mix}}(t-1)}^{k,\varepsilon}}(\tilde{X}_{t_{\text{mix}}t}^{k,\varepsilon} = y) \right) \\ &\leq \left(1 - \frac{\rho}{2M} \right)^{\lfloor m/t_{\text{mix}} \rfloor} \\ &\leq \exp \left(-\frac{\rho m}{2Mt_{\text{mix}}} \right) \leq \delta, \end{aligned}$$

as was to be shown. $\square$

B.3 Detailed Balance of Escape Probabilities

The following ‘detailed balance equation’ for hitting times, proved in Proposition 3.1 of [Betz and Le Roux \(2016\)](#) for nonreversible Markov chains, will be useful. We reproduce the proof here for convenience.

Proposition B.10. *For an irreducible, positive recurrent Markov chain X on a state space S with unique stationary distribution π , for all $x, y \in S$,*

$$\pi(x)\mathbb{P}_x(\bar{\tau}_y < \bar{\tau}_x) = \pi(y)\mathbb{P}_y(\bar{\tau}_x < \bar{\tau}_y).$$

Proof. For arbitrary $z \in S$, it holds that

$$\mathbb{E}_z[\bar{\tau}_x] = \mathbb{E}_z[\min\{\bar{\tau}_x, \bar{\tau}_y\}] + \mathbb{E}_z[(\bar{\tau}_x - \bar{\tau}_y)1_{\{\bar{\tau}_x > \bar{\tau}_y\}}] = \mathbb{E}_z[\min\{\bar{\tau}_x, \bar{\tau}_y\}] + \mathbb{E}_y[\bar{\tau}_x]\mathbb{P}_z(\bar{\tau}_x > \bar{\tau}_y). \quad (6)$$

Taking $z = y$ in (6) gives $\mathbb{E}_y[\min\{\bar{\tau}_x, \bar{\tau}_y\}] = \mathbb{E}_y[\bar{\tau}_x]\mathbb{P}_y(\bar{\tau}_x < \bar{\tau}_y)$, and substituting this in (6) with x, y swapped and $z = y$ yields

$$\frac{1}{\pi(y)} = \mathbb{E}_y[\bar{\tau}_y] = \mathbb{E}_y[\min\{\bar{\tau}_x, \bar{\tau}_y\}] + \mathbb{E}_x[\bar{\tau}_y]\mathbb{P}_y(\bar{\tau}_x < \bar{\tau}_y) = (\mathbb{E}_y[\bar{\tau}_x] + \mathbb{E}_x[\bar{\tau}_y])\mathbb{P}_y(\bar{\tau}_x < \bar{\tau}_y)$$

or

$$\pi(y)\mathbb{P}_y(\bar{\tau}_x < \bar{\tau}_y) = \frac{1}{\mathbb{E}_y[\bar{\tau}_x] + \mathbb{E}_x[\bar{\tau}_y]} = \pi(x)\mathbb{P}_x(\bar{\tau}_y < \bar{\tau}_x) \quad (7)$$

by symmetry. $\square$

C Perturbative Analysis of Metastable Dynamics

C.1 Quantitative Metastable Dynamics

We first show the following useful bound.

Lemma C.1. *There exists $\nu > 0$ such that for all $k \in [K]$ and distinct states $x, y \in C_k$, the unperturbed chain X^0 satisfies*

$$\mathbb{P}_x(\bar{\tau}_y^0 < \bar{\tau}_x^0) \geq \frac{\nu}{\log M}.$$

Proof. From (7) it holds that

$$\mu_k(y)\mathbb{P}_y(\bar{\tau}_x^0 < \bar{\tau}_y^0) = \frac{1}{\mathbb{E}_y[\bar{\tau}_x^0] + \mathbb{E}_x[\bar{\tau}_y^0]}. \quad (8)$$

By Assumption 1 it holds that $\mu_k(y) = \Theta(1/M)$. Moreover since the skipped chain $(X_{t_{\text{mix}}t}^0)_{t \geq 0}$ is well-mixed,

$$\mathbb{P}_x(\tau_y \geq m) \leq \sup \prod_{t=1}^{\lfloor m/t_{\text{mix}} \rfloor} \left(1 - \mathbb{P}_{X_{t_{\text{mix}}(t-1)}^0}(X_{t_{\text{mix}}t}^0 = m)\right) \leq \exp\left(-\frac{\rho m}{2Mt_{\text{mix}}}\right).$$

It follows that

$$\mathbb{E}_x[\bar{\tau}_y^0] = \sum_{m=0}^{\infty} \mathbb{P}_x(\tau_y \geq m) \leq \left(1 - \exp\left(-\frac{\rho}{2Mt_{\text{mix}}}\right)\right)^{-1} = O(M \log M)$$

and $\mathbb{E}_x[\bar{\tau}_y^0] = O(M \log M)$ by symmetry. The statement then follows from (8). $\square$

To study the cluster transition dynamics, we begin with a decomposition of hitting probabilities, which is closely related to the theory of stochastic complementation (Meyer, 1989). Here, we follow the proof in Betz and Le Roux (2016).

Lemma C.2. *For $C \subseteq S$, $x \in S$ and $y \in C$ such that $\mathbb{P}_x(\bar{\tau}_C^\varepsilon < \infty) = 1$, it holds that*

$$\mathbb{P}_x(X_{\bar{\tau}_C^\varepsilon}^\varepsilon = y) = p^\varepsilon(x, y) + \sum_{z \in C^c} \frac{\mathbb{P}_x(\bar{\tau}_z^\varepsilon < \bar{\tau}_C^\varepsilon)}{\mathbb{P}_z(\bar{\tau}_C^\varepsilon < \bar{\tau}_z^\varepsilon)} p^\varepsilon(z, y). \quad (9)$$

Proof. If $X_{\bar{\tau}_C^\varepsilon}^\varepsilon = y$, either X^ε has moved directly from x to y or has first moved to some $z = X_{\bar{\tau}_C^\varepsilon - 1}^\varepsilon \notin C$. Conditioning on the number of returns to z before transitioning to y yields

$$\begin{aligned} \mathbb{P}_x(X_{\bar{\tau}_C^\varepsilon}^\varepsilon = y) &= p^\varepsilon(x, y) + \sum_{z \in C^c} \sum_{n \geq 1} \mathbb{P}_x(\bar{\tau}_z^\varepsilon < \bar{\tau}_C^\varepsilon, X_{\bar{\tau}_z^\varepsilon + n}^\varepsilon = y) \\ &= p^\varepsilon(x, y) + \sum_{z \in C^c} \sum_{n \geq 1} \mathbb{P}_x(\bar{\tau}_z^\varepsilon < \bar{\tau}_C^\varepsilon) \mathbb{P}_z(\bar{\tau}_z^\varepsilon < \bar{\tau}_C^\varepsilon)^{n-1} p^\varepsilon(z, y) \\ &= p^\varepsilon(x, y) + \sum_{z \in C^c} \frac{\mathbb{P}_x(\bar{\tau}_z^\varepsilon < \bar{\tau}_C^\varepsilon)}{1 - \mathbb{P}_z(\bar{\tau}_z^\varepsilon < \bar{\tau}_C^\varepsilon)} p^\varepsilon(z, y), \end{aligned}$$

concluding (9). Note that $p^\varepsilon(x, y)$ must be added separately even for $z = x$ as the second term only counts returns to x for time $t > 0$. $\square$